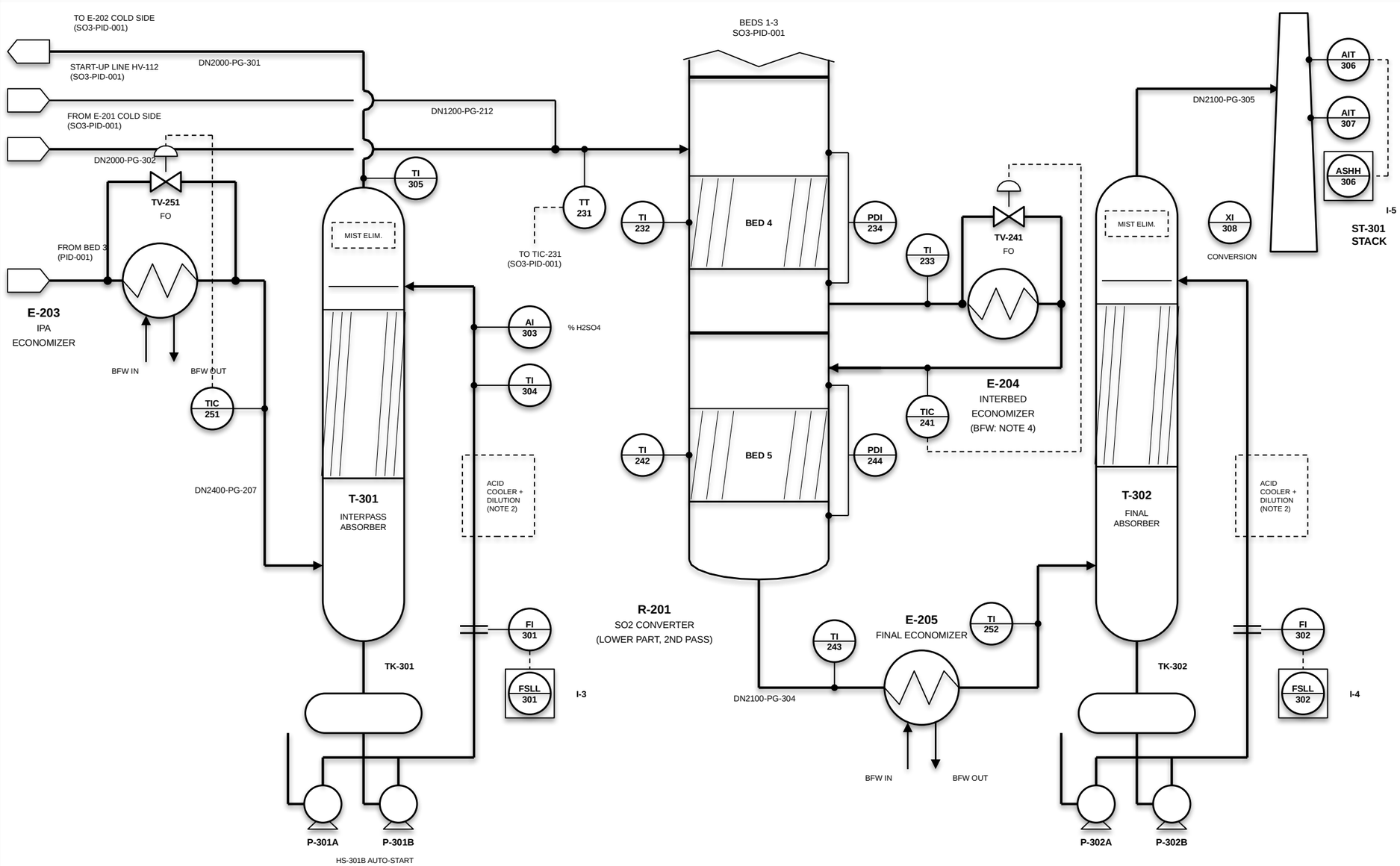


PRODUCT ACID, DILUTION WATER AND ACID CROSS-FLOWS: ACID SYSTEM P&ID (NOTE 2)

LEGEND

PROCESS LINE		UTILITY LINE	
SIGNAL LINE		LINE JUMP (NO JOIN)	
CONTROL VALVE FO/FC		SIS SHUT-OFF VALVE	
MANUAL VALVE (LC)		ORIFICE ELEMENT	
TI (DCS INSTRUMENT)		TI (FIELD INSTRUMENT)	
TI (SIS FUNCTION)		OFF-SHEET LINK	
CATALYST BED		S&T EXCHANGER	

CONTROL LOOP SCHEDULE

LOOP	MEASURES	MOVES	FAIL	SET / NOTE
FIC-103	SULFUR TO F-101	FV-103	FC	40.9 t/h (PRODUCTION)
FFC-101	AIR/SULFUR RATIO	FIC-101 SP	-	4.99 kNm3/t (14 % SO2)
FIC-101	COMBUSTION AIR	FV-101	FO	CAS, MIN 100 kNm3/h
AIC-104	SO2 INTO BED 1	FIC-102 SP	-	12.0 % SO2
FIC-102	DILUTION AIR	FV-102	FO	CAS FROM AIC-104
TIC-201	BED 1 INLET	TV-201 (B-101 BYP.)	FC	420 C
TIC-211	BED 2 INLET	TV-211 (E-201 BYP.)	FO	445 C
TIC-221	BED 3 INLET	TV-221 (E-202 BYP.)	FO	440 C
TIC-231	BED 4 INLET	TV-231 (SH-201 BYP.)	FO	425 C (NOTE 4)
TIC-241	BED 5 INLET	TV-241 (E-204 BYP.)	FO	428 C
TIC-251	GAS TO T-301	TV-251 (E-203 BYP.)	FO	190 C
HIC-109	-	H-101 FIRING	FC	START-UP ONLY

SIS INTERLOCKS (ACTION: SULFUR CUT)

I-	INITIATOR	SETTING	ACTION
1	FSLL-100 LOW TOTAL AIR	< 90 kNm3/h, 3 s	CLOSE XV-103
2	TSHH-203 BED 1 EXIT	> 660 C, 5 s	CLOSE XV-103
3	FSLL-301 IPA ACID FLOW	< 50 %, 10 s	CLOSE XV-103
4	FSLL-302 FAT ACID FLOW	< 50 %, 10 s	CLOSE XV-103
5	ASHH-306 STACK SO2	> 3000 ppm, 5 min *	CLOSE XV-103
6	HS-001 ESD PUSHBUTTON	MANUAL	XV-103 + STOP K-101, H-101

DESIGN POINT (OTS MODEL)

BED	IN C	OUT C	X %	PDI kPa
1	420	631	63.1	4.5
2	445	528	88.1	3.0
3	440	465	95.8	3.6
4	425	441	99.70	2.9
5	428	427	99.80	3.0

12.0 % SO2 / 9.0 % O2 INTO BED 1; STACK 286 ppm SO2;
PI-200 35 kPa g; K-101 DISCHARGE 60 kPa g.

GENERAL NOTES:

1. SEE SO3-PID-001 NOTES 1-3 FOR DESIGN BASIS, LINE SIZING AND SIS PHILOSOPHY. RETURN GAS FROM T-301 IS REHEATED IN E-202 THEN E-201 (SO3-PID-001) BEFORE BED 4.
2. T-301 / T-302 ACID CIRCULATION: PUMP TANKS, ACID COOLERS, DILUTION WATER AND PRODUCT ACID CONTROLS ARE ON THE ACID SYSTEM P&ID - SHOWN HERE AS PACKAGE LIMITS.
3. FSLL-301 ALSO AUTO-STARTS SPARE P-301B (HS-301B ARMED); P-302B IS STARTED MANUALLY. LOSS OF ACID CIRCULATION > 10 s TRIPS SULFUR (I-3, I-4).
4. E-203/E-204/E-205 ECONOMIZERS: BFW SIDE ON STEAM SYSTEM P&ID. E-204 BYPASS (TV-241) SETS BED 5 INLET; E-203 BYPASS (TV-251) KEEPS GAS TO T-301 ABOVE THE ACID DEW POINT.
5. AIT-306 / AIT-307 STACK ANALYZERS: SO2 (ppmv, DRY) AND SO3/H2SO4 MIST (mg/Nm3). XI-308 OVERALL CONVERSION CALCULATED IN DCS FROM AI-104, FI-100 AND AIT-306.
6. BED TEMPERATURES AND CONVERSIONS ARE THE OTS DESIGN POINT (SO3 CONVERTER VER 1); PLANT VALUES TBC.

REV	DESCRIPTION	DATE
A	ISSUED FOR OTS DEVELOPMENT (REVIEW)	25-SEP-26

CHULA CHEM. ENG. - SO3 CONVERTER VER 1 (OTS)

P&ID - INTERPASS ABSORPTION, BEDS 4-5, FINAL ABSORPTION

DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 25-SEP-26
DWG NO. SO3-PID-002	REV A	SCALE N.T.S.	SHEET 2 OF 2